PCBC H Parametrics Practice Test A Name _____ _____/12

Non-Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

_____

_____

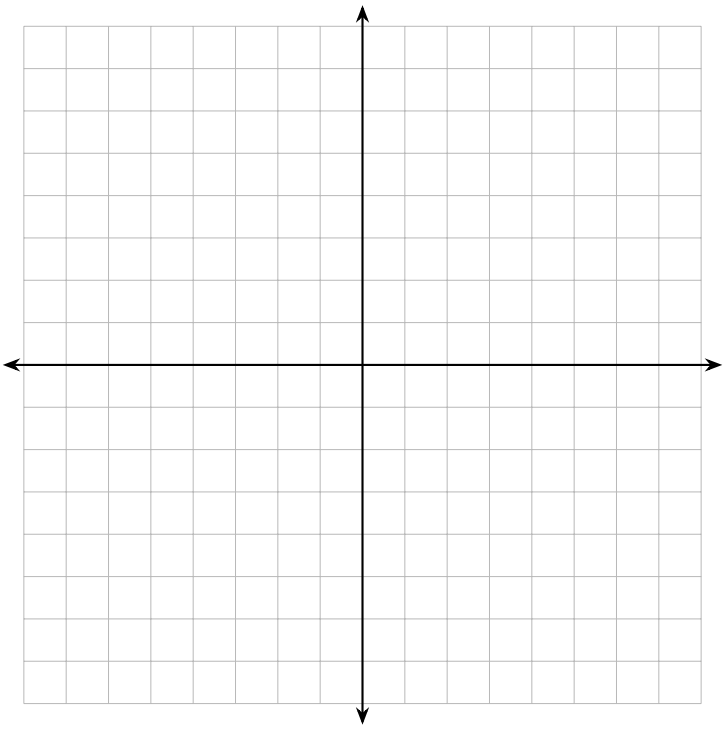
Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: 3 points per question is a practice split of the posted 12 points.

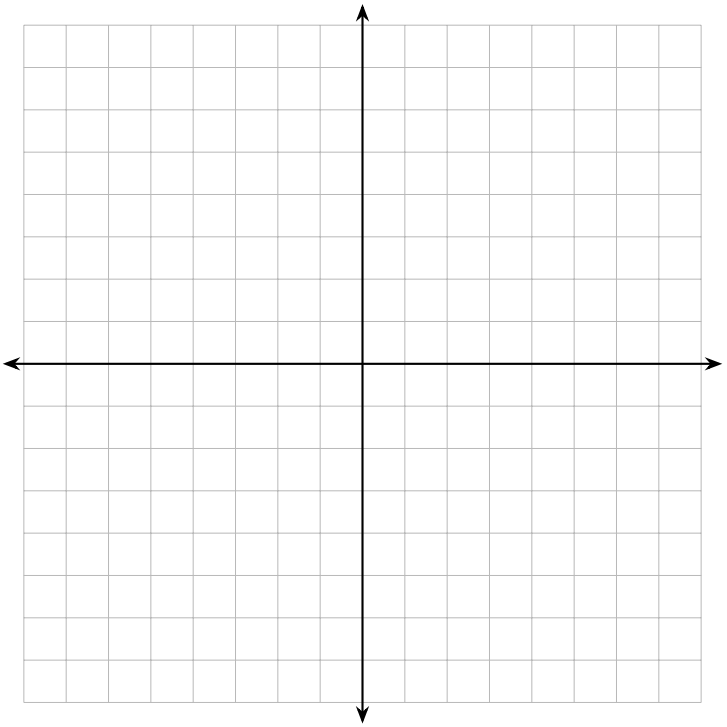
Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

1. $x = t^2 - 2, \quad y = t + 1, \quad$ for t in $[-2, 2]$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

2. $x = 3\cos t, \quad y = 2\sin t, \quad \text{for } t \text{ in } [0, \pi]$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

3. $y = x^2 - 4x + 1$ **3 pts**

Representation 1 *Representation 2*

Calculator | 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth.
(practice instructions)

4. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet. *(Formulas printed here as a practice choice.)*

A baseball is hit from a height of 3 ft at an angle of inclination of 35° with a velocity of 110 ft/sec. Write the parametric equations that model its path. For how many seconds is the baseball in flight? What horizontal distance does it travel? **3 pts**

PCBC H Parametrics Practice Test A: Answer Key

12 points

Every value was checked in Python (sympy and 30-digit arithmetic). Work is on the left, the reason for each line is on the right.

---

1. $x = t^2 - 2,\ y = t + 1,$ for t in $[-2, 2]$ *Non-Calculator*

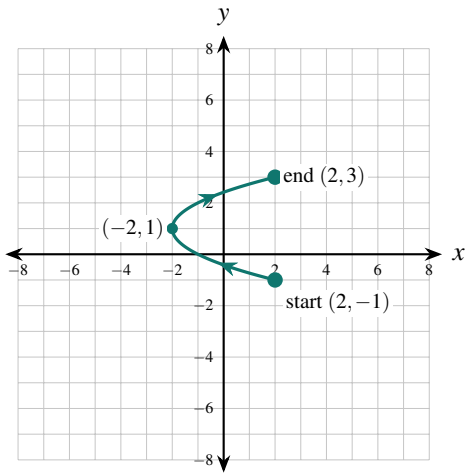
$\boxed{x = (y-1)^2 - 2, \quad -1 \le y \le 3 \ \ (\text{so } -2 \le x \le 2)}$

Graph: sideways parabola from $(2,-1)$ through the vertex $(-2,1)$ to $(2,3)$, traced bottom to top.

Work	Why
$y - 1 = t$	y gives t in one step.
$x = (y-1)^2 - 2$	Substitute $t = y - 1$ into x.
$t = -2 \Rightarrow (2,-1);\ t = 2 \Rightarrow (2,3)$	Both endpoints give $x = 2$.
$t = 0 \Rightarrow (-2,1)$	$x = t^2 - 2$ turns around at $t = 0$: smallest x.
$-1 \le y \le 3,\ -2 \le x \le 2$	$y = t + 1$ only increases, so y runs -1 to 3. The x endpoints alone (both 2) would miss $x = -2$.
solid dots at $(2,-1)$ and $(2,3)$	Closed interval, two different end points.

t	(x, y)
-2	$(2,-1)$
-1	$(-1,0)$
0	$(-2,1)$
1	$(-1,2)$
2	$(2,3)$

y goes up the whole time, so the arrows point up along the curve.



Practice scoring (3 pts): 1 for the equation, 1 for the restriction ($-1 \le y \le 3$, or $-2 \le x \le 2$ together with the y bound), 1 for the graph with end dots and upward arrows.

Trap: stating only $x = 2$ from the two endpoints. x turns around at $t = 0$, so it reaches -2. The whole parabola $x = (y-1)^2 - 2$ keeps going past $y = 3$; this curve stops there.

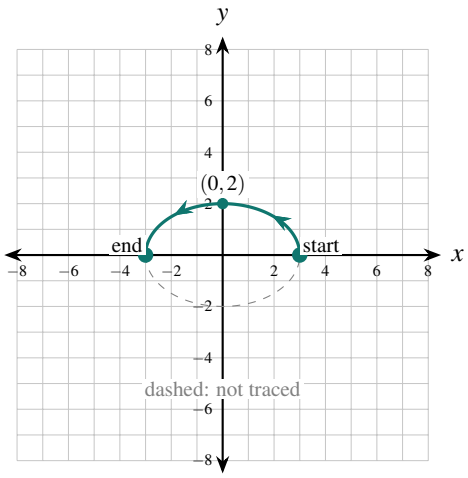
2. $x = 3\cos t,\ y = 2\sin t,$ for t in $[0, \pi]$ *Non-Calculator*

$\boxed{\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1, \quad y \geq 0}$ (top half of the ellipse)
From $(3,0)$ counterclockwise to $(-3,0)$; $-3 \leq x \leq 3$, $0 \leq y \leq 2$.

Work	Why
$\cos t = \dfrac{x}{3}, \quad \sin t = \dfrac{y}{2}$	Isolate each trig function.
$\cos^2 t + \sin^2 t = 1$	Pythagorean identity.
$\left(\dfrac{x}{3}\right)^2 + \left(\dfrac{y}{2}\right)^2 = 1$	Substitute.
$\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$	Ellipse, center $(0,0)$, 3 left and right, 2 up and down.
$\sin t \geq 0$ on $[0, \pi]$, so $y \geq 0$	Only the top half is traced.
$t = 0 : (3,0);\ t = \frac{\pi}{2} : (0,2);\ t = \pi : (-3,0)$	Right, then top, then left: counterclockwise.

t	(x,y)
0	$(3,0)$
$\frac{\pi}{2}$	$(0,2)$
π	$(-3,0)$

Closed interval with two different end points: solid dots at $(3,0)$ and $(-3,0)$.



Practice scoring (3 pts): 1 for $\frac{x^2}{9} + \frac{y^2}{4} = 1$, 1 for keeping only $y \geq 0$, 1 for the half-ellipse graph with end dots and counterclockwise arrows.

Trap: drawing the whole ellipse. t stops at π, so the bottom half is never traced.

3. $y = x^2 - 4x + 1$ *Non-Calculator*

$\boxed{x = t,\ y = t^2 - 4t + 1,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t + 2,\ y = t^2 - 3,\ \text{for } t \text{ in } (-\infty, \infty)}$

Work	**Why**
Representation 1: $x = t$	Let x be the parameter.
$y = t^2 - 4t + 1,\ \ t \text{ in } (-\infty, \infty)$	The parabola has every real x, so t is every real number.
Representation 2: $y = (x^2 - 4x + 4) - 4 + 1$	Complete the square: half of -4 is -2, $(-2)^2 = 4$.
$y = (x - 2)^2 - 3$	Vertex form, vertex $(2, -3)$.
$t = x - 2 \ \Rightarrow\ x = t + 2$	Let t be the inside of the square.
$y = t^2 - 3,\ \ t \text{ in } (-\infty, \infty)$	Substitute.
Check: $(t+2)^2 - 4(t+2) + 1 = t^2 + 4t + 4 - 4t - 8 + 1 = t^2 - 3$	Put $x = t + 2$ back into the original.

Practice scoring (3 pts): 1 for each correct pair of equations, 1 for the t interval on both. Any other correct pair also counts.

Trap: $t = x - 2$ means $x = t + 2$, not $t - 2$. Write "for t in $(-\infty, \infty)$" with both.

4. Baseball launched from $h_0 = 3$ ft at $\theta = 35°$ with $v_0 = 110$ ft/sec. *Calculator*

$\boxed{x = (110\cos 35°)t, \quad y = -16t^2 + (110\sin 35°)t + 3}$
In flight $\boxed{\approx 3.990 \text{ sec}}$, travels $\boxed{\approx 359.555 \text{ ft}}$ horizontally.

Work	Why
$x = (110\cos 35°)t \approx 90.107t$	Horizontal: $v_0 \cos\theta$ times t.
$y = -16t^2 + (110\sin 35°)t + 3 \approx -16t^2 + 63.093t + 3$	Vertical: gravity, $v_0 \sin\theta$ times t, plus the starting height.
$0 = -16t^2 + (110\sin 35°)t + 3$	It lands when $y = 0$.
$t = \dfrac{-110\sin 35° \pm \sqrt{(110\sin 35°)^2 - 4(-16)(3)}}{2(-16)}$	Quadratic formula with $a = -16$, $b = 110\sin 35°$, $c = 3$.
$t = \dfrac{-63.093 \pm \sqrt{4172.778}}{-32}$	$b^2 \approx 3980.778$ and $-4ac = +192$.
$t \approx 3.990$ or $t \approx -0.047$	Reject the negative time.
$x = (110\cos 35°)(3.99032\ldots)$	Put the unrounded t into x (keep it stored).
$x \approx 359.555$ ft	Round only at the end.

Calculator setup (TI-Nspire), and a graph check

- Angle mode: **Degree**, or type the ° symbol after the angle. In radian mode every number is wrong.
- Graph check: Graphs page, Menu, Graph Entry/Edit, Parametric. $x1(t) = 110\cos(35°)t$, $y1(t) = -16t^2 + 110\sin(35°)t + 3$, $0 \le t \le 4.2$. Window x: 0 to 400, y: -5 to 75. Trace near $t = 3.990$: y crosses 0 with $x \approx 359.5$.
- Not asked: the top of the path is at $t = \frac{110\sin 35°}{32} \approx 1.972$ sec, height ≈ 65.200 ft.

Practice scoring (3 pts): 1 for both parametric equations, 1 for the flight time, 1 for the horizontal distance.

Trap: rounding t to 3.99 before finding x gives ≈ 359.526 ft, which is off at the thousandths. Keep the full t.